

Lab: Linear Regression

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```
# adding libraries by loading the MASS AND ISLR2 Packages
```

```
library(MASS)
```

```
library(ISLR2)
```

```
##
```

```
## Attaching package: 'ISLR2'
```

```
## The following object is masked from 'package:MASS':
```

```
##
```

```
## Boston
```

```
#Showing first 10 rows of Boston dataset retrieved via ISLR2 Package
```

```
head(Boston)
```

```
##      crim zn  indus chas   nox   rm  age   dis rad tax ptratio lstat medv
## 1 0.00632 18  2.31    0 0.538 6.575 65.2 4.0900   1 296    15.3  4.98 24.0
## 2 0.02731  0  7.07    0 0.469 6.421 78.9 4.9671   2 242    17.8  9.14 21.6
## 3 0.02729  0  7.07    0 0.469 7.185 61.1 4.9671   2 242    17.8  4.03 34.7
## 4 0.03237  0  2.18    0 0.458 6.998 45.8 6.0622   3 222    18.7  2.94 33.4
## 5 0.06905  0  2.18    0 0.458 7.147 54.2 6.0622   3 222    18.7  5.33 36.2
## 6 0.02985  0  2.18    0 0.458 6.430 58.7 6.0622   3 222    18.7  5.21 28.7
```

```
# After specify a dataset
```

```
lm.fit <- lm(medv ~ lstat, data = Boston)
```

```
attach(Boston)
```

```
lm.fit <- lm(medv ~ lstat)
```

```
#Display output on model
```

```
lm.fit # Basic information about model output
```

```
##
```

```
## Call:
```

```
## lm(formula = medv ~ lstat)
```

```
##
```

```
## Coefficients:
```

```
## (Intercept)      lstat
```

```
##      34.55      -0.95
```

```
summary(lm.fit) #Summary of the basic information like p-value,....
```

```
##  
## Call:  
## lm(formula = medv ~ lstat)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -15.168  -3.990  -1.318   2.034  24.500   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept) 34.55384    0.56263   61.41  <2e-16 ***   
## lstat       -0.95005    0.03873  -24.53  <2e-16 ***   
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 6.216 on 504 degrees of freedom  
## Multiple R-squared:  0.5441, Adjusted R-squared:  0.5432   
## F-statistic: 601.6 on 1 and 504 DF,  p-value: < 2.2e-16
```

```
#Display names of information on model
```

```
names(lm.fit) #Display names of information on model
```

```
## [1] "coefficients" "residuals"      "effects"      "rank"  
## [5] "fitted.values" "assign"        "qr"           "df.residual"  
## [9] "xlevels"      "call"         "terms"        "model"
```

```
coef(lm.fit) #extraction
```

```
## (Intercept)      lstat  
## 34.5538409 -0.9500494
```

```
confint(lm.fit) #confidential interval
```

```
##              2.5 %      97.5 %  
## (Intercept) 33.448457 35.6592247  
## lstat       -1.026148 -0.8739505
```

```
#Confidential interval and Predict
```

```
predict(lm.fit , data.frame(lstat = (c(5, 10, 15))),  
interval = "confidence")
```

```
##      fit      lwr      upr  
## 1 29.80359 29.00741 30.59978  
## 2 25.05335 24.47413 25.63256  
## 3 20.30310 19.73159 20.87461
```

```
#Predict
```

```
predict(lm.fit , data.frame(lstat = (c(5, 10, 15))),  
interval = "prediction")
```

```
##          fit          lwr          upr  
## 1 29.80359 17.565675 42.04151  
## 2 25.05335 12.827626 37.27907  
## 3 20.30310  8.077742 32.52846
```

```
# Plot median value (medv) vs percentage of lower-status population (lstat)  
plot(lstat, medv)
```

```
# Add the fitted linear regression line to the plot
```

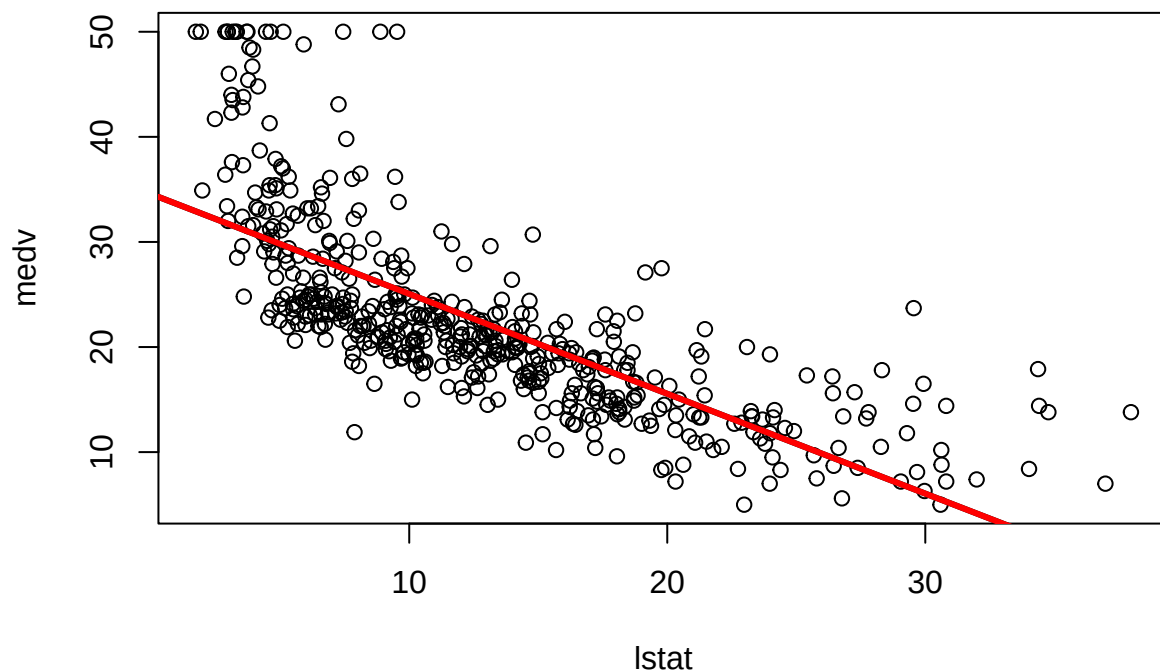
```
abline(lm.fit)
```

```
# Add the regression line with line width 3
```

```
abline(lm.fit, lwd = 3)
```

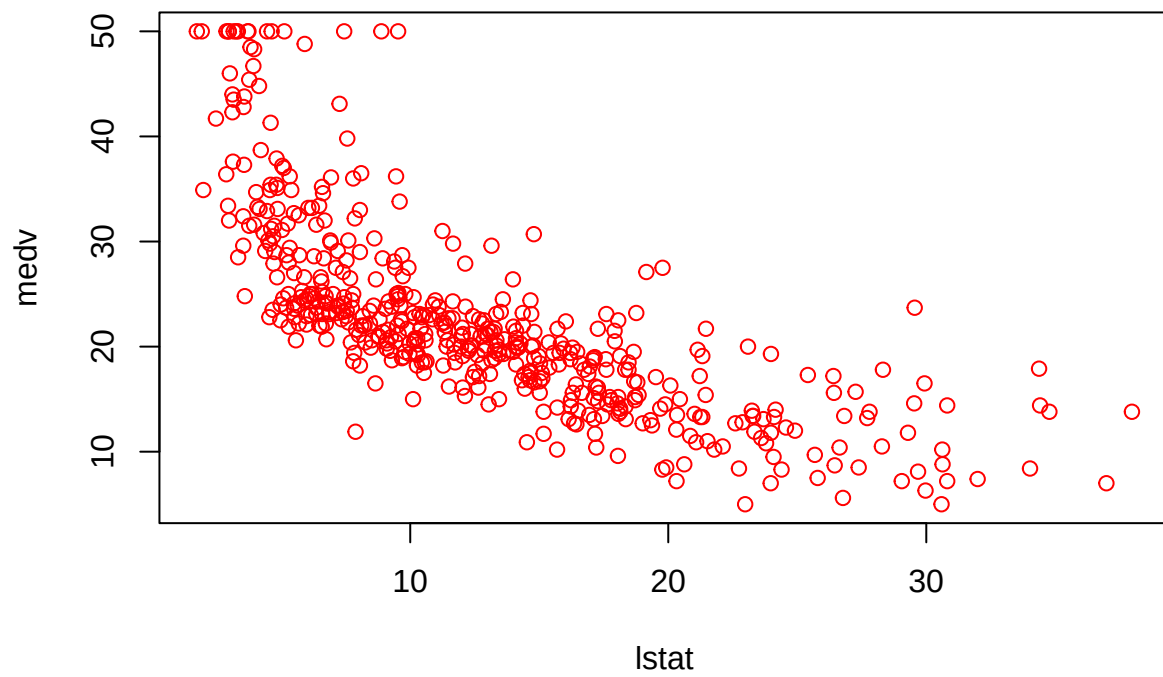
```
# Add the regression line with line width 3 and red color
```

```
abline(lm.fit, lwd = 3, col = "red")
```

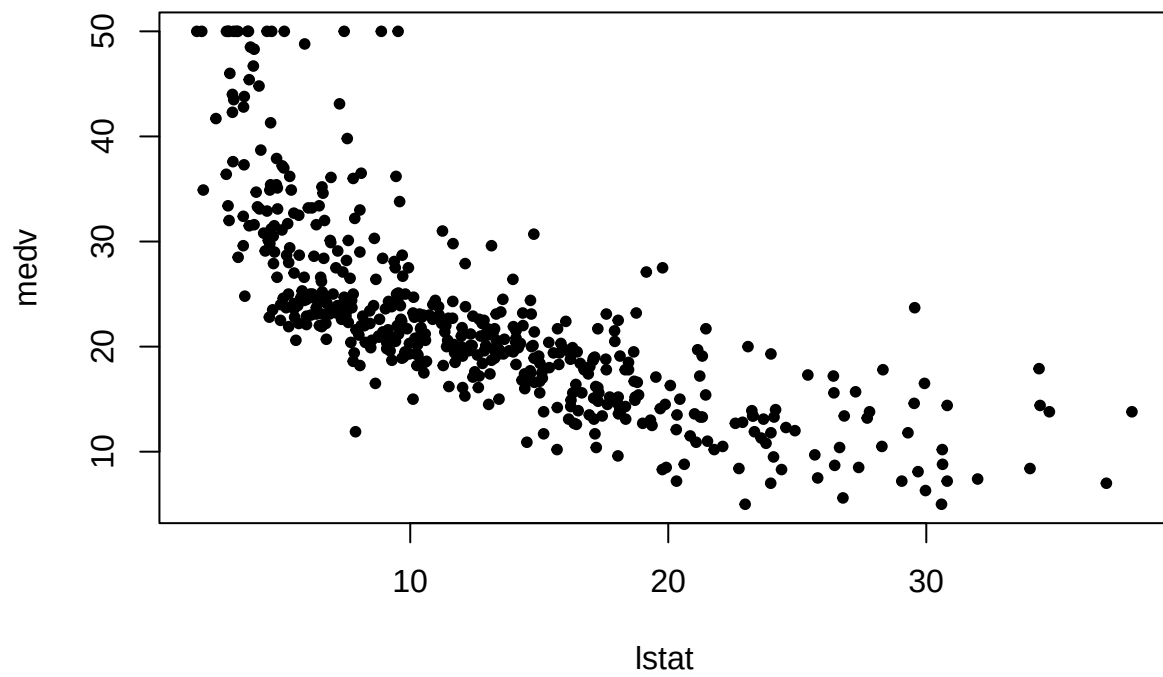


```
# Create a scatter plot with the points displayed in red
```

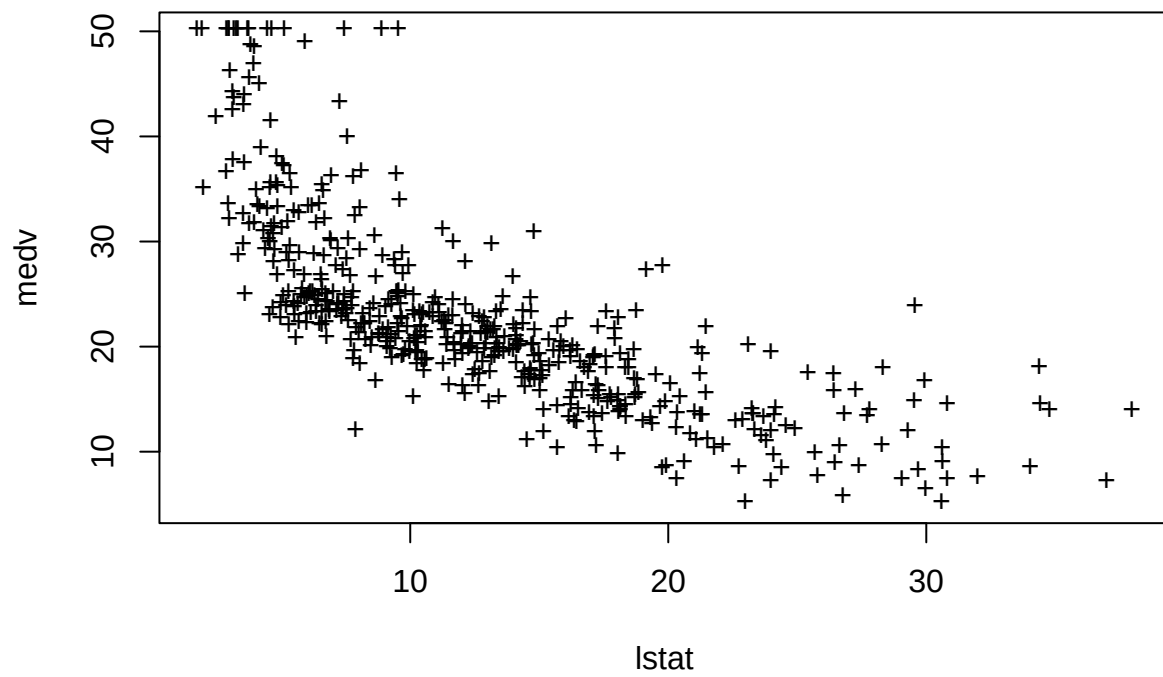
```
plot(lstat, medv, col = "red")
```



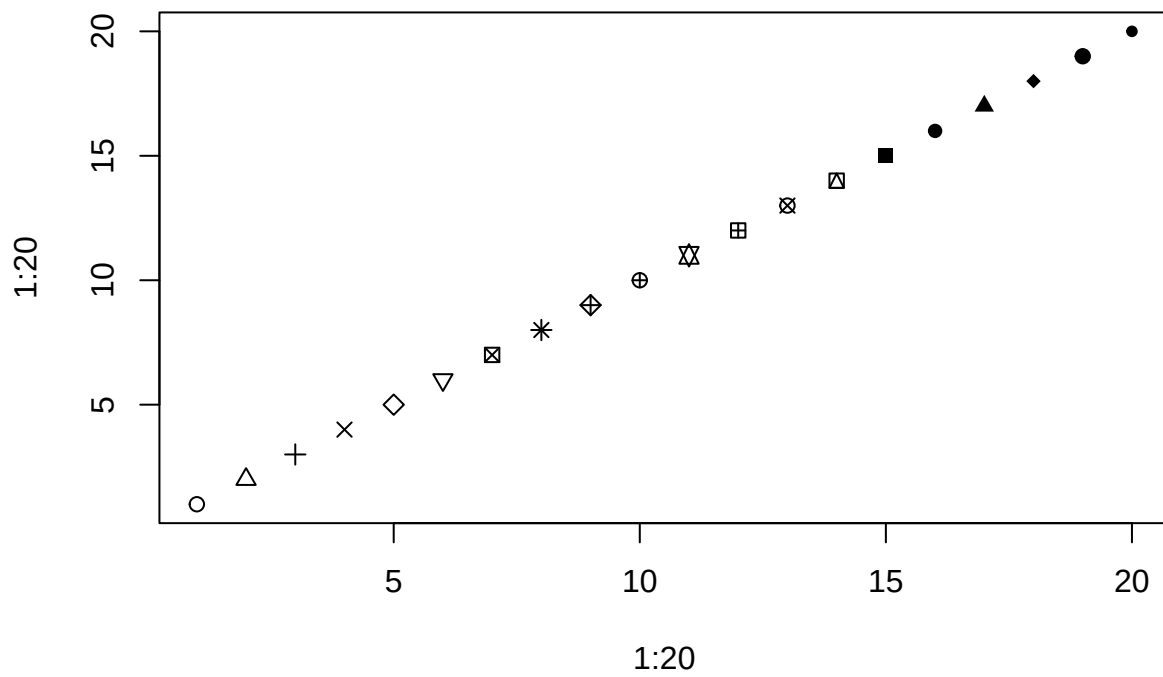
```
# Create a scatter plot using solid circle points (pch = 20)  
plot(lstat, medv, pch = 20)
```



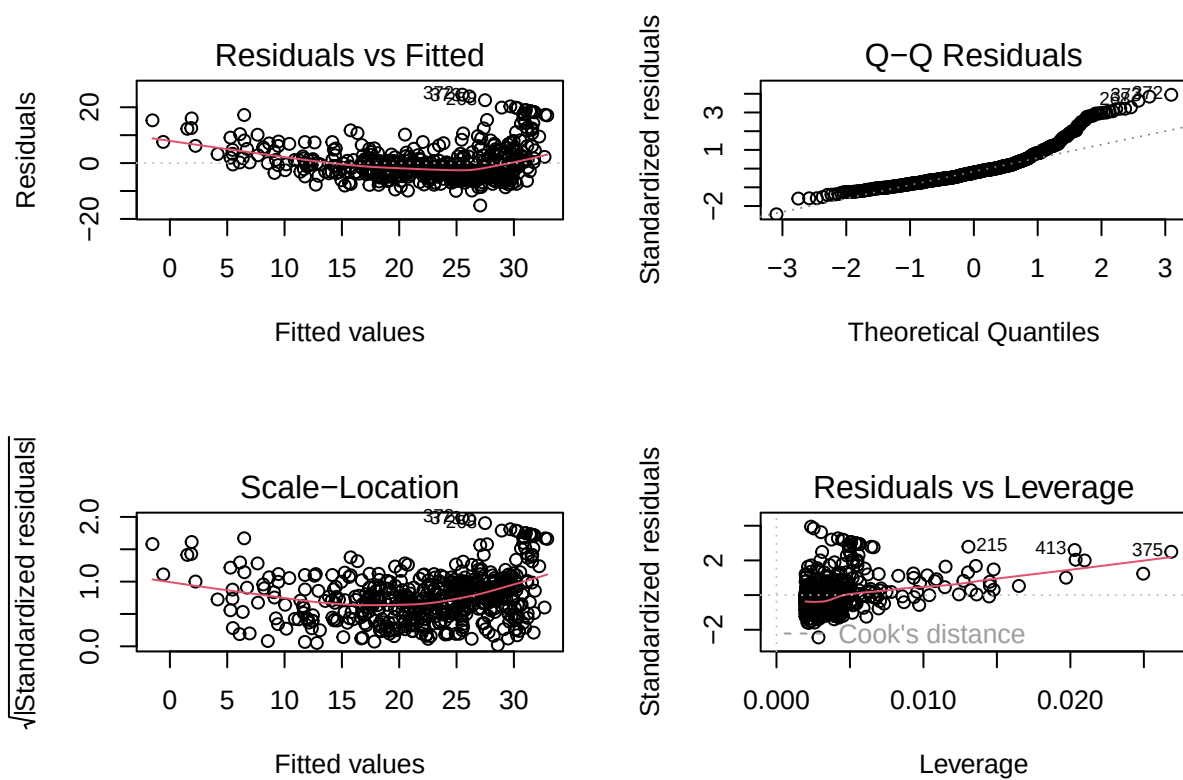
```
# Create a scatter plot using "+" symbols for the points  
plot(lstat, medv, pch = "+")
```



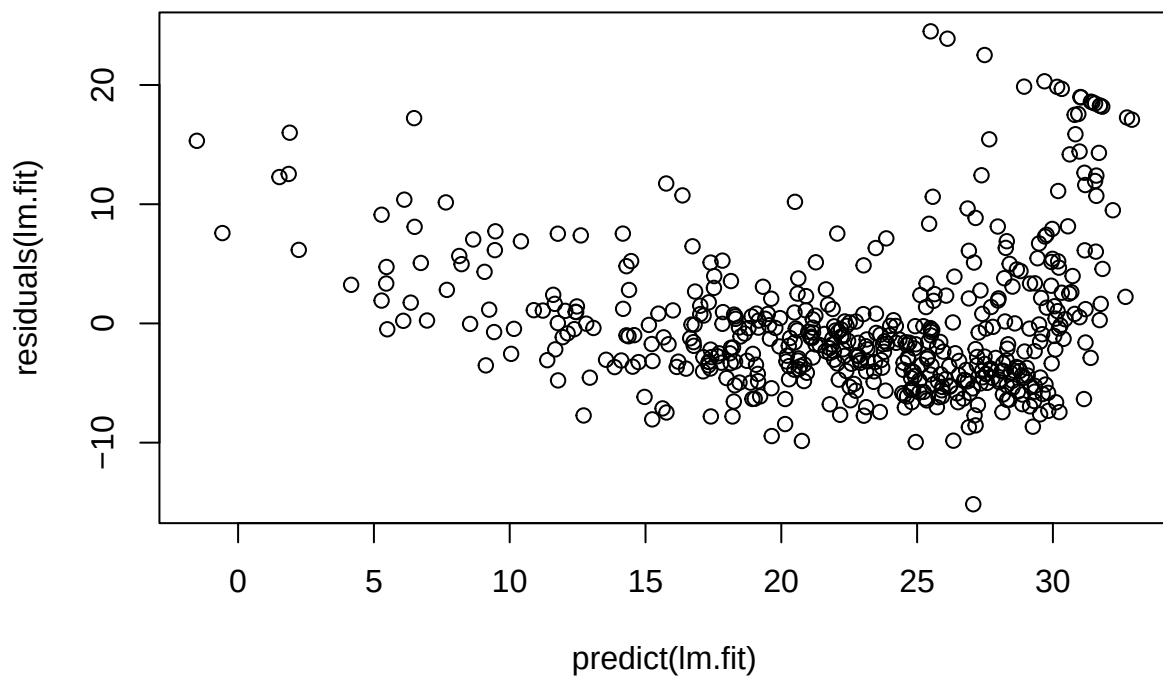
```
# Display different plotting symbols (pch 1 to 20)
plot(1:20, 1:20, pch = 1:20)
```



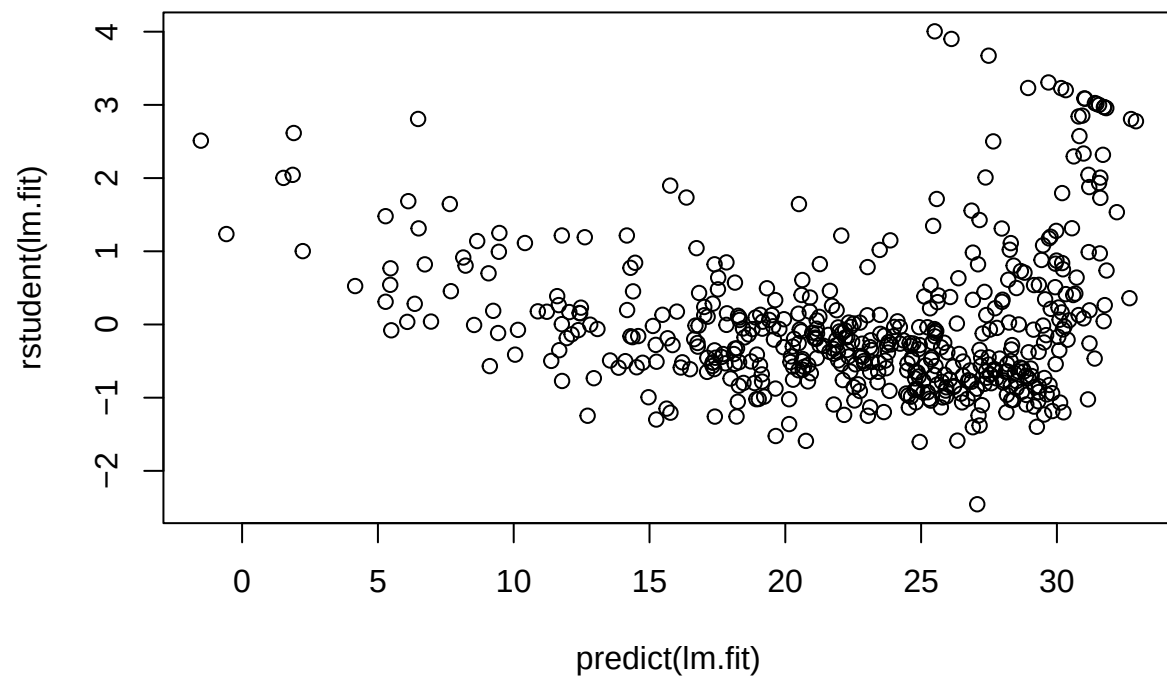
```
# Arrange the plotting area into 2 rows and 2 columns  
par(mfrow = c(2, 2))  
  
# Display the four diagnostic plots for the linear regression model  
plot(lm.fit)
```



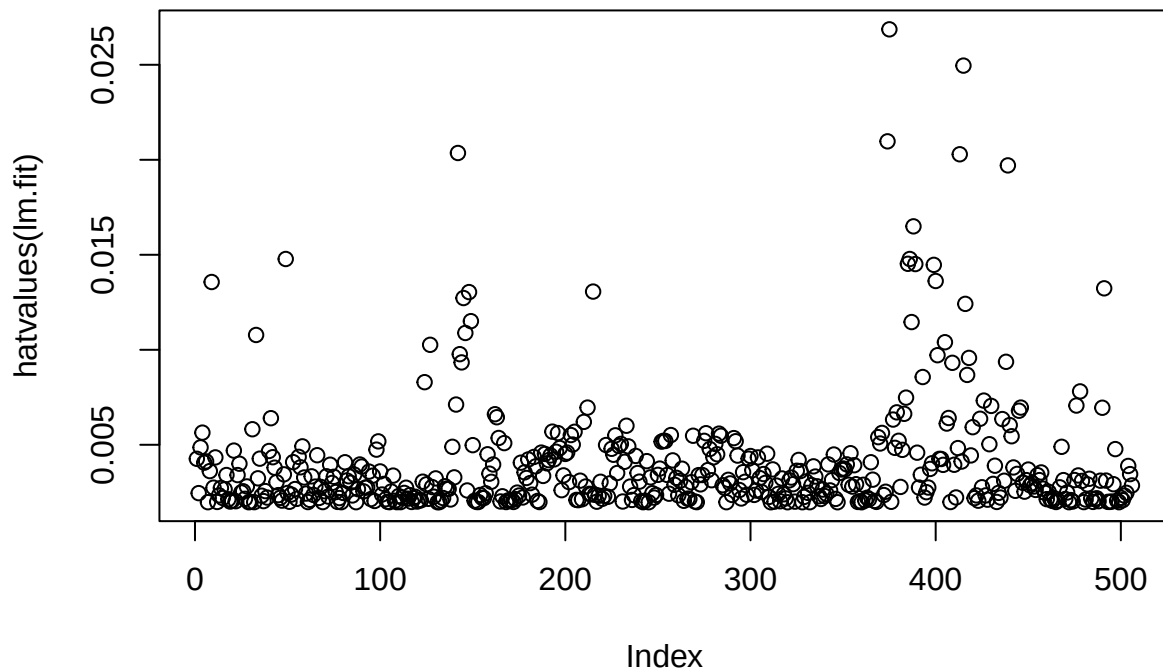
```
# Plot the predicted values against the model residuals
plot(predict(lm.fit), residuals(lm.fit))
```



```
# Plot the predicted values against the studentized residuals  
plot(predict(lm.fit), rstudent(lm.fit))
```



```
# Plot the leverage (hat) values for each observation  
plot(hatvalues(lm.fit))
```



```
# Find the observation with the largest leverage value
which.max(hatvalues(lm.fit))
```

```
## 375
## 375
```

```
# Fit a multiple linear regression model using lstat and age to predict medv
lm.fit <- lm(medv ~ lstat + age, data = Boston)
```

```
# Display the summary of the fitted regression model
summary(lm.fit)
```

```
##
## Call:
## lm(formula = medv ~ lstat + age, data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.981  -3.978  -1.283   1.968   23.158
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  33.22276    0.73085  45.458  < 2e-16 ***
## lstat       -1.03207    0.04819 -21.416  < 2e-16 ***
## age          0.03454    0.01223   2.826  0.00491 **
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.173 on 503 degrees of freedom
## Multiple R-squared:  0.5513, Adjusted R-squared:  0.5495
## F-statistic:   309 on 2 and 503 DF,  p-value: < 2.2e-16

# Fit a multiple linear regression model using all variables in the Boston dataset
lm.fit <- lm(medv ~ ., data = Boston)

# Display the summary of the fitted regression model
summary(lm.fit)
```

```
##
## Call:
## lm(formula = medv ~ ., data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1304  -2.7673  -0.5814   1.9414  26.2526
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  41.617270   4.936039   8.431 3.79e-16 ***
## crim        -0.121389   0.033000  -3.678 0.000261 ***
## zn           0.046963   0.013879   3.384 0.000772 ***
## indus        0.013468   0.062145   0.217 0.828520
## chas         2.839993   0.870007   3.264 0.001173 **
## nox        -18.758022   3.851355  -4.870 1.50e-06 ***
## rm           3.658119   0.420246   8.705 < 2e-16 ***
## age          0.003611   0.013329   0.271 0.786595
## dis         -1.490754   0.201623  -7.394 6.17e-13 ***
## rad          0.289405   0.066908   4.325 1.84e-05 ***
## tax         -0.012682   0.003801  -3.337 0.000912 ***
## ptratio     -0.937533   0.132206  -7.091 4.63e-12 ***
## lstat       -0.552019   0.050659 -10.897 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.798 on 493 degrees of freedom
## Multiple R-squared:  0.7343, Adjusted R-squared:  0.7278
## F-statistic: 113.5 on 12 and 493 DF,  p-value: < 2.2e-16
```

```
# Load the car package, which provides functions for regression diagnostics
library(car)
```

```
## Loading required package: carData
```

```
# Calculate the Variance Inflation Factor (VIF) for each predictor
vif(lm.fit)
```

```
##      crim      zn      indus      chas      nox      rm      age      dis
```

```
## 1.767486 2.298459 3.987181 1.071168 4.369093 1.912532 3.088232 3.954037
##      rad      tax ptratio  lstat
## 7.445301 9.002158 1.797060 2.870777
```

```
# Fit a multiple linear regression model using all predictors except age
lm.fit1 <- lm(medv ~ . - age, data = Boston)
```

```
# Display the summary of the fitted regression model
summary(lm.fit1)
```

```
##
## Call:
## lm(formula = medv ~ . - age, data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1851  -2.7330  -0.6116   1.8555  26.3838
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  41.525128   4.919684   8.441 3.52e-16 ***
## crim        -0.121426   0.032969  -3.683 0.000256 ***
## zn           0.046512   0.013766   3.379 0.000785 ***
## indus        0.013451   0.062086   0.217 0.828577
## chas         2.852773   0.867912   3.287 0.001085 **
## nox        -18.485070   3.713714  -4.978 8.91e-07 ***
## rm           3.681070   0.411230   8.951 < 2e-16 ***
## dis         -1.506777   0.192570  -7.825 3.12e-14 ***
## rad           0.287940   0.066627   4.322 1.87e-05 ***
## tax          -0.012653   0.003796  -3.333 0.000923 ***
## ptratio      -0.934649   0.131653  -7.099 4.39e-12 ***
## lstat        -0.547409   0.047669 -11.483 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.794 on 494 degrees of freedom
## Multiple R-squared:  0.7343, Adjusted R-squared:  0.7284
## F-statistic: 124.1 on 11 and 494 DF, p-value: < 2.2e-16
```

```
# Update the existing model by removing the age variable
lm.fit1 <- update(lm.fit, ~ . - age)
```

```
# Display the summary of the updated model
summary(lm.fit1)
```

```
##
## Call:
## lm(formula = medv ~ crim + zn + indus + chas + nox + rm + dis +
##      rad + tax + ptratio + lstat, data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1851  -2.7330  -0.6116   1.8555  26.3838
```

```
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  41.525128   4.919684   8.441 3.52e-16 ***
## crim        -0.121426   0.032969  -3.683 0.000256 ***
## zn          0.046512   0.013766   3.379 0.000785 ***
## indus       0.013451   0.062086   0.217 0.828577
## chas        2.852773   0.867912   3.287 0.001085 **
## nox       -18.485070   3.713714  -4.978 8.91e-07 ***
## rm          3.681070   0.411230   8.951 < 2e-16 ***
## dis       -1.506777   0.192570  -7.825 3.12e-14 ***
## rad         0.287940   0.066627   4.322 1.87e-05 ***
## tax       -0.012653   0.003796  -3.333 0.000923 ***
## ptratio    -0.934649   0.131653  -7.099 4.39e-12 ***
## lstat      -0.547409   0.047669 -11.483 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.794 on 494 degrees of freedom
## Multiple R-squared:  0.7343, Adjusted R-squared:  0.7284
## F-statistic: 124.1 on 11 and 494 DF, p-value: < 2.2e-16
```

```
# Fit a regression model including lstat, age, and their interaction
summary(lm(medv ~ lstat * age, data = Boston))
```

```
##
## Call:
## lm(formula = medv ~ lstat * age, data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.806  -4.045  -1.333   2.085  27.552
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) 36.0885359  1.4698355  24.553 < 2e-16 ***
## lstat       -1.3921168  0.1674555  -8.313 8.78e-16 ***
## age         -0.0007209  0.0198792  -0.036  0.9711
## lstat:age    0.0041560  0.0018518   2.244  0.0252 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.149 on 502 degrees of freedom
## Multiple R-squared:  0.5557, Adjusted R-squared:  0.5531
## F-statistic: 209.3 on 3 and 502 DF, p-value: < 2.2e-16
```

```
# Fit a quadratic regression model using lstat and lstat squared
lm.fit2 <- lm(medv ~ lstat + I(lstat^2), data = Boston)
```

```
# Display the summary of the quadratic regression model
summary(lm.fit2)
```

```
##
```

```
## Call:
## lm(formula = medv ~ lstat + I(lstat^2), data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.2834  -3.8313  -0.5295   2.3095  25.4148
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 42.862007   0.872084   49.15  <2e-16 ***
## lstat       -2.332821   0.123803  -18.84  <2e-16 ***
## I(lstat^2)   0.043547   0.003745   11.63  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.524 on 503 degrees of freedom
## Multiple R-squared:  0.6407, Adjusted R-squared:  0.6393
## F-statistic: 448.5 on 2 and 503 DF,  p-value: < 2.2e-16
```

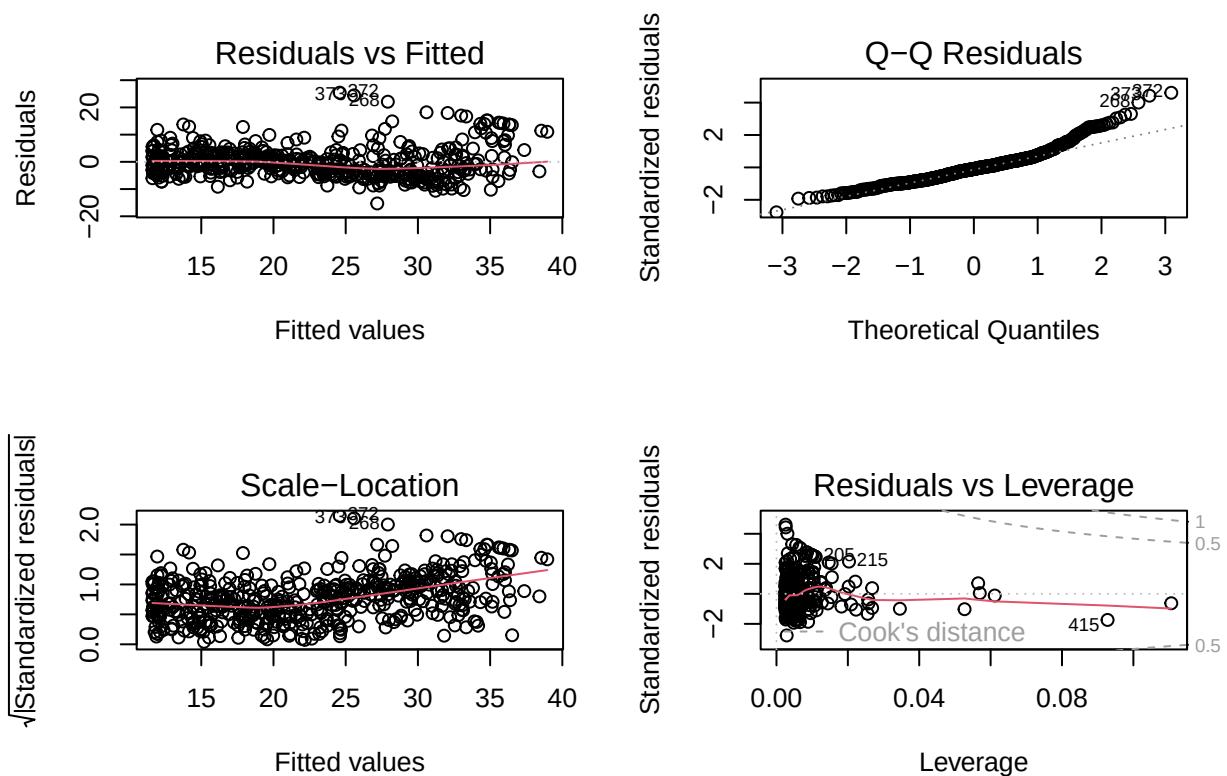
```
# Fit the simple linear regression model
lm.fit <- lm(medv ~ lstat, data = Boston)

# Compare the linear model with the quadratic model using ANOVA
anova(lm.fit, lm.fit2)
```

```
## Analysis of Variance Table
##
## Model 1: medv ~ lstat
## Model 2: medv ~ lstat + I(lstat^2)
##   Res.Df  RSS Df Sum of Sq    F    Pr(>F)
## 1     504 19472
## 2     503 15347   1    4125.1 135.2 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Arrange the plotting area into 2 rows and 2 columns
par(mfrow = c(2, 2))

# Generate four diagnostic plots for the quadratic regression model
plot(lm.fit2)
```



```
# Fit a polynomial regression model of degree 5
lm.fit5 <- lm(medv ~ poly(lstat, 5), data = Boston)

# Display the summary of the fifth-degree polynomial model
summary(lm.fit5)

##
## Call:
## lm(formula = medv ~ poly(lstat, 5), data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.5433  -3.1039  -0.7052   2.0844  27.1153
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    22.5328     0.2318  97.197 < 2e-16 ***
## poly(lstat, 5)1 -152.4595     5.2148 -29.236 < 2e-16 ***
## poly(lstat, 5)2   64.2272     5.2148  12.316 < 2e-16 ***
## poly(lstat, 5)3  -27.0511     5.2148  -5.187 3.10e-07 ***
## poly(lstat, 5)4   25.4517     5.2148   4.881 1.42e-06 ***
## poly(lstat, 5)5  -19.2524     5.2148  -3.692 0.000247 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.215 on 500 degrees of freedom
```

```
## Multiple R-squared:  0.6817, Adjusted R-squared:  0.6785
## F-statistic: 214.2 on 5 and 500 DF,  p-value: < 2.2e-16
```

```
# Fit a regression model using the logarithm of rm as the predictor
summary(lm(medv ~ log(rm), data = Boston))
```

```
##
## Call:
## lm(formula = medv ~ log(rm), data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.487  -2.875  -0.104   2.837  39.816
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -76.488      5.028  -15.21  <2e-16 ***
## log(rm)       54.055      2.739   19.73  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.915 on 504 degrees of freedom
## Multiple R-squared:  0.4358, Adjusted R-squared:  0.4347
## F-statistic: 389.3 on 1 and 504 DF,  p-value: < 2.2e-16
```

```
# Display the first six rows of the Carseats dataset
head(Carseats)# Fit a multiple linear regression model using all predictors
```

```
##   Sales CompPrice Income Advertising Population Price ShelveLoc Age Education
## 1  9.50      138     73          11         276    120      Bad    42         17
## 2 11.22      111     48          16         260     83     Good    65         10
## 3 10.06      113     35          10         269     80   Medium    59         12
## 4  7.40      117    100           4         466     97   Medium    55         14
## 5  4.15      141     64           3         340    128     Bad    38         13
## 6 10.81      124    113          13         501     72     Bad    78         16
##   Urban  US
## 1   Yes Yes
## 2   Yes Yes
## 3   Yes Yes
## 4   Yes Yes
## 5   Yes  No
## 6    No Yes
```

```
# and include the interactions between Income and Advertising,
# and between Price and Age
lm.fit <- lm(Sales ~ . + Income:Advertising + Price:Age,
             data = Carseats)
```

```
# Display the summary of the fitted regression model
summary(lm.fit)
```

```
##
```

```
## Call:
## lm(formula = Sales ~ . + Income:Advertising + Price:Age, data = Carseats)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.9208 -0.7503  0.0177  0.6754  3.3413
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    6.5755654   1.0087470    6.519 2.22e-10 ***
## CompPrice      0.0929371   0.0041183   22.567 < 2e-16 ***
## Income         0.0108940   0.0026044    4.183 3.57e-05 ***
## Advertising    0.0702462   0.0226091    3.107 0.002030 **
## Population     0.0001592   0.0003679    0.433 0.665330
## Price        -0.1008064   0.0074399  -13.549 < 2e-16 ***
## ShelveLocGood  4.8486762   0.1528378   31.724 < 2e-16 ***
## ShelveLocMedium 1.9532620   0.1257682   15.531 < 2e-16 ***
## Age          -0.0579466   0.0159506   -3.633 0.000318 ***
## Education     -0.0208525   0.0196131   -1.063 0.288361
## UrbanYes       0.1401597   0.1124019    1.247 0.213171
## USYes         -0.1575571   0.1489234   -1.058 0.290729
## Income:Advertising 0.0007510  0.0002784    2.698 0.007290 **
## Price:Age      0.0001068   0.0001333    0.801 0.423812
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.011 on 386 degrees of freedom
## Multiple R-squared:  0.8761, Adjusted R-squared:  0.8719
## F-statistic: 210 on 13 and 386 DF, p-value: < 2.2e-16
```

```
# Attach the Carseats dataset so its variables can be accessed directly
attach(Carseats)

# Display the contrast coding used for the categorical variable ShelveLoc
contrasts(ShelveLoc)
```

```
##      Good Medium
## Bad      0      0
## Good     1      0
## Medium   0      1
```

Writing Functions

```
LoadLibraries <- function() {
  library ( ISLR2 )
  library (MASS)
  print (" The libraries have been loaded .")
}
```

```
LoadLibraries ()
```

```
## [1] " The libraries have been loaded ."
```